

Business Requirements Specification

University Course Feedback Analytics Platform

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Project Name: University Course Feedback Analytics Platform

Executive Summary

The University Course Feedback Analytics Platform is a comprehensive system designed to collect, analyze, and visualize student feedback data to improve teaching quality and student learning outcomes. The system will enable stakeholders at all levels—students, faculty, and administrators—to access actionable insights from feedback data in real time.

Primary Objective: Establish a data-driven culture where feedback directly informs course improvements, faculty development, and strategic resource allocation.

Key Benefits:

- Faculty gain real-time visibility into student satisfaction and can respond promptly to concerns
- Administrators identify high-performing and struggling courses for targeted support
- Students access anonymized feedback to make informed course selection decisions
- Department heads track trends across courses to optimize curriculum and staffing
- University demonstrates commitment to continuous improvement in teaching quality

Scope: Complete feedback collection, storage, analysis, and visualization system with role-based dashboards and automated sentiment analysis.

1. Stakeholder Analysis

1.1 Primary Stakeholders

Students (Feedback Providers & Consumers)

- Role: Submit feedback via surveys; view anonymized feedback on course pages
- Goals: Provide honest feedback; make informed course choices; see improvements happen
- Pain Points: Survey fatigue; concerns about anonymity; no feedback on whether their input was heard
- Success Metric: 60%+ response rate on feedback surveys; increased course selections based on feedback

Faculty (Feedback Recipients)

- Role: Receive feedback; respond to student concerns; track teaching improvements
- Goals: Improve their teaching; address specific student concerns; demonstrate continuous improvement
- Pain Points: Bulk feedback emails (no prioritization); difficult to track trends; slow feedback loops
- Success Metric: Faculty satisfaction with feedback system; reduced time to respond (target ≤ 7 days)

Department Administrators (Monitors & Reporters)

- Role: Oversee feedback system; generate reports; allocate professional development resources
- Goals: Identify struggling courses; celebrate high performers; make data-driven staffing decisions
- Pain Points: Manual data compilation; lack of real-time visibility; spotting trends is time-consuming
- Success Metric: Ability to generate semester reports in < 1 hour; alerts on critical issues

Faculty Leadership/Deans (Strategic Decision Makers)

- Role: Review high-level trends; inform curriculum decisions; allocate resources
- Goals: Optimize teaching quality across the institution; support faculty development; track progress
- Pain Points: Aggregate data hard to obtain; can't drill down to specific issues; strategic planning is slow
- Success Metric: Using system insights to inform curriculum and hiring decisions

1.2 Secondary Stakeholders

IT Department: System support, database management, security compliance

Quality Assurance: Data integrity verification, system reliability monitoring

Student Services: Access to feedback data for student support initiatives

2. User Personas

Persona 1: Dr. Sarah Chen (Faculty Member)

Characteristics: Mid-career professor, teaching 3 courses per semester, 15 years experience

Goals: Understand how students perceive her teaching; make targeted improvements; document professional growth

Pain Points: Gets bulk feedback via email; hard to spot patterns; unsure which feedback is most critical

System Needs: Easy-to-read dashboard showing my course ratings; latest student comments; actionable insights; trending data (semester-over-semester)

Success: Identifies 2-3 specific improvements to implement next semester; responds to student concerns within 7 days

Persona 2: Alex Rodriguez (Student)

Characteristics: Junior, taking 4 courses, first time using feedback system

Goals: Provide honest feedback; see that feedback leads to improvements; make informed course choices

Pain Points: Previous surveys felt like a dead end; unsure if feedback is anonymous; too many questions

System Needs: Simple rating + comment interface; confirmation that feedback is anonymous; ability to view feedback on course pages

Success: Submits 2-3 feedback entries per semester; sees improvements implemented by faculty

Persona 3: Dr. James Mitchell (Department Chair)

Characteristics: Senior faculty, managing 12 courses and 18 faculty; 20 years experience

Goals: Ensure teaching quality; identify struggling courses early; allocate professional development resources

Pain Points: Manual compilation of feedback takes 10+ hours per semester; can't see trends across courses; reports are outdated by the time he gets them

System Needs: Automated reports; system-wide metrics; alerts on critical issues; drill-down to course/faculty level

Success: Generates semester report in <1 hour; identifies struggling courses within 2 weeks; allocates PD resources proactively

Persona 4: Maya Patel (Dean of Academic Affairs)

Characteristics: Senior administrator, managing 4 departments, 150+ courses; strategic planning focus

Goals: Optimize teaching quality institution-wide; inform curriculum decisions; support strategic initiatives

Pain Points: High-level data aggregation is manual; can't track progress over years; competitive analysis not available

System Needs: Institution-wide dashboards; multi-year trend analysis; alerts on strategic issues; export capabilities for presentations

Success: Uses system insights in strategic planning; tracks KPIs quarterly; demonstrates improvement to accreditation bodies

3. Functional Requirements

3.1 Feedback Collection & Submission

FR1.1: Feedback Survey Interface

- Students can access feedback surveys via course page, email link, or mobile app
- Survey includes: 1-5 star rating scale, 4-5 Likert-scale questions, open comment field
- Survey should take <5 minutes to complete
- Accessible on web and mobile devices

FR1.2: Anonymous Submission

- Student name/ID NOT collected during submission
- Submission IP address is logged but not displayed to faculty
- System guarantees anonymity unless student voluntarily identifies themselves in comment

FR1.3: Feedback Validation

- Rating required (1-5 scale); comments optional
- System rejects submissions with invalid ratings
- System validates comment length (min 10 chars, max 2000 chars)
- Duplicate submissions by same student within 24 hours flagged as potential duplicates

3.2 Feedback Analysis & Storage

FR2.1: Data Persistence

- All feedback stored in database with timestamp, course, semester, rating, comment
- Data retained for 7+ years for historical analysis
- Soft delete capability for erroneous submissions

FR2.2: Sentiment Analysis

- System automatically analyzes comment sentiment (positive/negative/neutral)
- Sentiment score stored with each comment (confidence 0-100%)
- Topic extraction: identify key topics mentioned (e.g., "lecture pacing", "instructor clarity")
- Updated in real-time as new feedback arrives

FR2.3: Data Quality Assurance

- System flags potential spam/inappropriate comments (manual review)
- Automated deduplication identifies near-duplicate comments
- Null value handling: missing ratings handled consistently
- Data validation: all ratings in range [1, 5]

3.3 Dashboard & Reporting

FR3.1: Faculty Dashboard

- Faculty views own course metrics (avg rating, sentiment distribution, response rate)
- Latest feedback comments displayed with sentiment tags
- Trend visualization (semester-over-semester comparison)
- Alert on critical feedback or low satisfaction

FR3.2: Administrator Dashboard

- System-wide metrics: total feedback collected, avg satisfaction, sentiment distribution
- Course ranking: top 5 and bottom 5 courses by satisfaction
- Heatmap: courses × metrics matrix, color-coded by performance
- Alert panel: courses below thresholds flagged for intervention

FR3.3: Student Course Page

- Students can view anonymized feedback summary on course page
- Shows: avg rating (1-5), sentiment breakdown (%), top 3 feedback themes
- Does NOT show individual comments (privacy protection)
- Helps students make informed course selection

FR3.4: Report Generation

- Admins can generate semester reports (PDF/Excel)
- Reports include: overall metrics, course rankings, trends, recommendations
- Exportable data: raw feedback data (for further analysis), aggregated metrics
- Report generation <5 minutes for 100+ courses

3.4 User Access & Authentication

FR4.1: Role-Based Access Control

- Student role: submit feedback, view anonymized course summaries
- Faculty role: view own course feedback, respond to comments, track improvements
- Admin role: view all feedback, generate reports, manage users
- System enforces role-based visibility (faculty cannot see other faculty's feedback)

FR4.2: Authentication

- Login via university SSO (single sign-on)
- Multi-factor authentication optional for admins
- Session timeout after 30 minutes of inactivity

3.5 Notifications & Alerts

FR5.1: Critical Alerts

- Automated alert when course satisfaction drops below 3.5/5.0
- Alert when comment sentiment is strongly negative (confidence >80%)
- Alert when response rate <40% (low engagement)
- Alerts sent to faculty and department chair within 24 hours

FR5.2: Feedback Notifications

- Faculty notified of new feedback (daily digest, not real-time spam)
- Student receives confirmation of feedback submission
- Faculty prompted to respond to feedback (after 3 days with no response)

4. Non-Functional Requirements

4.1 Performance

NFR1.1: Response Time

- Dashboard load time: <2 seconds on standard bandwidth
- API response time: <500ms for 95th percentile
- Report generation: <5 minutes for full semester data

NFR1.2: Scalability

- Support 500+ concurrent users
- Handle 50,000+ feedback submissions per semester
- System performance maintained with 10× data growth

4.2 Reliability & Availability

NFR2.1: Uptime

- 99.5% availability (target 4 hours downtime per year)
- Peak usage periods (week before semester end): zero unplanned downtime

NFR2.2: Data Backup

- Nightly backup of all data to off-site storage
- Recovery time objective (RTO): <4 hours
- Recovery point objective (RPO): <1 hour

4.3 Security & Privacy

NFR3.1: Data Privacy

- Student identifiers NOT stored with feedback (anonymity guaranteed)
- Faculty access restricted to their own courses only
- Admin access logged and auditable
- Compliance with FERPA (Family Educational Rights and Privacy Act)

NFR3.2: Data Encryption

- All data transmitted via HTTPS (TLS 1.2+)

- Database encryption at rest (AES-256)
- Sensitive data (passwords, tokens) hashed with bcrypt

NFR3.3: Audit Trail

- All access to dashboard logged (who, when, what data viewed)
- Changes to feedback flagged and logged
- Monthly audit report for compliance

4.4 Usability

NFR4.1: User Interface

- Intuitive navigation; no training required for students/faculty
- Mobile-responsive design; works on phones/tablets
- Accessibility: WCAG 2.1 Level AA compliance
- Support for 3+ languages (English, Spanish, Mandarin)

NFR4.2: Documentation

- User guides for each role (student, faculty, admin)
- Video tutorials for key workflows
- FAQ section addressing common questions

4.5 Maintainability

NFR5.1: Code Quality

- Open-source stack (avoid vendor lock-in)
- Comprehensive code documentation
- Automated testing (80%+ code coverage)

NFR5.2: Monitoring

- Real-time system health monitoring (CPU, memory, disk)
- Error logging and alerting
- Monthly performance reports

5. Key Performance Indicators (KPIs)

KPI	Definition	Target	Alert Threshold
Course Satisfaction	Avg rating (1-5) per course	≥ 4.0	< 3.5
Content Clarity	Avg rating on "Content was clear" question	≥ 4.0	< 3.5
Instructor Engagement	Avg rating on "Instructor was engaging" question	≥ 4.0	< 3.5
Sentiment Score	% of feedback with positive sentiment	$\geq 75\%$	$< 60\%$
Response Rate	(Students who submitted feedback / Enrolled students) $\times 100$	$\geq 60\%$	$< 40\%$
Feedback Velocity	Avg days for faculty to respond to feedback	≤ 7 days	> 14 days
System Uptime	Availability of feedback platform	99.5%	$< 99\%$
Data Quality	% of feedback records with valid ratings	100%	$< 99\%$

6. Assumptions & Constraints

6.1 Assumptions

- University has active SSO system for authentication
- Internet connectivity available to all stakeholders
- Faculty have willingness to implement feedback-based improvements
- Student feedback is honest and constructive
- Department leadership will support system adoption

6.2 Constraints

- Budget: System must operate within IT department budget allocation
- Timeline: Live pilot in Spring 2026, full rollout Fall 2026
- Technology: Must integrate with existing university systems (Blackboard, Banner)
- Data: Cannot mandate participation (feedback is voluntary)
- Staffing: One full-time support person allocated for system management

7. Out of Scope

The following are explicitly NOT included in this project:

- **Academic Management:** Grade recording, transcript management, course registration
- **External Integration:** Posting feedback to public platforms; auto-posting to social media
- **Advanced ML:** Predictive modeling, anomaly detection beyond simple thresholds
- **Mobile App:** Native iOS/Android app (web-responsive design sufficient for Phase 1)
- **Multilingual Support:** System English-only in Phase 1; internationalization in Phase 2
- **Third-Party Analytics:** Integration with external analytics platforms (future enhancement)

8. Success Criteria

The project will be considered successful when:

1. **Adoption:** 60%+ of faculty actively using the system; 50%+ of eligible students submitting feedback
2. **Data Quality:** 95%+ of feedback submissions have valid ratings; sentiment analysis accuracy >85%
3. **Performance:** Dashboard loads <2 seconds; 99.5% system uptime during pilot
4. **Impact:** Faculty report using feedback insights to make course improvements; measurable KPI improvements in 3+ courses
5. **Satisfaction:** Faculty/admin Net Promoter Score (NPS) >50 for system usability
6. **Operations:** System operations require <5 hours/week management effort after launch

The End